

SECTION 13 34 20

PREFABRICATED STEEL BUILDINGS
12/24

PART 1 GENERAL

1.1 SECTION INCLUDES

Prefabricated Steel Buildings

1.2 RELATED SECTIONS

- a. Section 03 30 00 CAST-IN-PLACE CONCRETE
- b. Section 05 50 13 MISCELLANEOUS METAL FABRICATIONS
- c. Section 08 71 00 DOOR HARDWARE
- d. Section 07 92 00 JOINT SEALANTS
- e. Section 22 00 00 PLUMBING, GENERAL PURPOSE
- f. Division 26 - Electrical

1.3 REFERENCES

The publications listed below form a part of this specification to the extent referenced. The publications are referred to within the text by the basic designation only.

ASTM INTERNATIONAL (ASTM)

ASTM A513/A513M	(2024) Standard Specification for Electric-Resistance-Welded Carbon and Alloy Steel Mechanical Tubing
ASTM A653/A653M	(2023) Standard Specification for Steel Sheet, Zinc-Coated (Galvanized) or Zinc-Iron Alloy-Coated (Galvannealed) by the Hot-Dip Process
ASTM A1008/A1008M	(2023) Standard Specification for Steel, Sheet, Cold-Rolled, Carbon, Structural, High-Strength Low-Alloy, High-Strength Low-Alloy with Improved Formability, Solution Hardened, and Bake Hardenable
ASTM B209	(2014) Standard Specification for Aluminum and Aluminum-Alloy Sheet and Plate
ASTM B221	(2021) Standard Specification for Aluminum and Aluminum-Alloy Extruded Bars, Rods, Wire, Profiles, and Tubes
ASTM B632/B632M	(2024) Standard Specification for Aluminum-Alloy Rolled Tread Plate

ASTM C578 (2023) Standard Specification for Rigid,
Cellular Polystyrene Thermal Insulation

U.S. DEPARTMENT OF DEFENSE (DOD)

UFC 3-301-01 (2023; with Change 1, 2023) Structural
Engineering

1.4 DESIGN REQUIREMENTS

- a. Provide factory built, prefabricated structures capable of withstanding the effects of gravity loads and the following loads and stresses within limits and under conditions indicated.
- b. Wind Loads: Determine loads based on the following minimum design wind pressures:
 - (1) Wind Load: Buildings: Wind speed and exposure as indicated.
 - (2) Snow Loads: As indicated.
- c. Seismic Performance: Provide factory built, prefabricated structures and shelters capable of withstanding the effects of earthquake motions determined according to:
 - (1) UFC 3-301-01 Structural Engineering.
- d. Thermal Movements: Provide factory built, prefabricated structures and shelters that allow for thermal movements resulting from the following maximum change (range) in ambient and surface temperatures by preventing buckling, opening of joints, overstressing of components, failure of joint sealants, failure of connections, and other detrimental effects. Base engineering calculation on surface temperatures of materials due to both solar heat gain and nighttime-sky heat loss.
 - (1) Temperature Change (Range): 120 deg F, ambient; 180 deg F, material surfaces.
- e. Electrical Devices: Devices UL listed with wiring bearing UL classification and conforming to the current NEC.

1.5 PERFORMANCE REQUIREMENTS

Cooperate with regulatory agency or authority and provide data as requested by authority having jurisdiction.

1.6 SUBMITTALS

Government approval is required for submittals with a "G" designation; submittals not having a "G" designation are for Contractor Quality Control approval. Submit the following in accordance with Section 01 33 00 SUBMITTAL PROCEDURES:

SD-02 Shop Drawings

Plans; G

SD-03 Product Data

Construction Details; G

Preparation Instructions; G

Storage and Handling Requirements; G

Installation Methods; G

SD-04 Samples

Color Chips; G

SD-07 Certificates

Product Certificates; G

SD-11 Closeout Submittals

Warranty; G

1.7 Submittal Lists

- a. Product Data: Manufacturer's data sheets on each product to be used, including:
 - (1) Construction details, material descriptions, dimensions of individual components and profiles, and finishes.
 - (2) Preparation instructions and recommendations.
 - (3) Storage and handling requirements and recommendations.
 - (4) Installation methods.
- b. Shop Drawings: Include plans, elevations, sections, details, and attachments to other work.
- c. Selection Samples: For each finish product specified, two complete sets of color chips representing manufacturer's full range of available colors and patterns.
- d. Certificates: Product certificates signed by the manufacturer certifying material compliance with specified performance characteristics and criteria, and physical requirements.

1.8 QUALITY ASSURANCE

Manufacturer Qualifications: Company specializing in manufacturing prefabricated structures and shelters with a minimum documented experience of twenty years and with a quality assurance program utilizing an independent third party quality control firm with a 5-stage, 35 step, quality inspection for each system.

Pre-installation Meetings: Conduct meetings to verify project requirements, substrate conditions, utility connections, manufacturer's installation instructions, and warranty requirements. Comply with Division 1 requirements.

1.9 DELIVERY, STORAGE, AND HANDLING

Store products in manufacturer's unopened packaging until ready for installation.

Protect all components and accessories from corrosion, deformation, damage and deterioration when stored at job site. Keep materials free from dirt and foreign matter.

1.10 PROJECT CONDITIONS

Maintain environmental conditions (temperature, humidity, and ventilation) within limits recommended by manufacturer for optimum results. Do not install products under environmental conditions outside manufacturer's absolute limits.

1.11 WARRANTY

Panel Built, Inc. warrants the building system purchased from Panel Built for a period of one (1) year from the date of shipment. This includes all materials and freight to rectify any product defects. Panel Built further warrants the Panel Built system (walls and roof) for a period of five (5) years against delamination, structural defects or system breakdown under normal wear and tear conditions. All buildings must be installed in accordance with Panel Built installation instructions, drawings and specifications.

Provide manufacturer's warranties on all accessory items provided such as, but not limited to, air conditioning, lights, and heating units.

PART 2 PRODUCTS

2.1 MATERIALS

- a. Aluminum: Alloy and temper recommended by manufacturer for type of use and finish specified, and as follows:
 - (1) Sheet: ASTM B209.
 - (2) Extruded Shapes: ASTM B221
 - (3) Rolled Tread Plate: ASTM B632/B632M, Alloy 6061-T4 or 6061-T6.
- b. Cold-Rolled Steel Sheet: ASTM A1008/A1008M, Commercial Steel (CS), Type B.
- c. Zinc-Coated (Galvanized) Steel Sheet: ASTM A653/A653M, commercial quality, G90 (Z275) coating designation; mill phosphatized.
- d. Aluminum Tread plate: 1/8-inch aluminum plate conforming to ASTM B209.
- e. Steel Mechanical Tubing: ASTM A513/A513M, welded steel mechanical tubing.
- f. Expanded Polystyrene (EPS) Core: Minimum of .95 pcf complying with ASTM C578 Type 1.
- g. Oriented Strand Board (OSB): Standard Grade; minimum physical properties conforming to APA PRP-108.
- h. Anchorages: Anchor bolts, as specified in Section 03 30 00.

2.2 PREFABRICATED STEEL BUILDINGS

- a. Size: 8'-6" x 13'-6" clear inside floor plan dimensions.
- b. Height: Nominal outside height 102 inches. Interior floor to ceiling height 96 inches.
- c. Prefabricated with 14 gauge mechanical tube, welded framing system. Exterior smooth and flat with no exposed fasteners on building exterior.
 - (1) Lifting lugs shall be provided.
 - (2) Forklift pockets shall be provided.
 - (3) Optional Decorative Reveal.
 - (4) Exterior custom veneer panel.
- d. Doors: Swinging door(s).
- e. Roof Type: Standing Seam Hip.
- f. Roof Overhang: 4 inch overhand (standard).
- g. Finish: Finish shall have a prime to paint system manufacturer's epoxy primer with a urethane finish as follows:
 - (1) Color as selected from manufacturers standard colors.
- h. Base/Floor. Floor is mounted to 2 inch to 4 inch minimum tube/channel steel base frame. Finished floor of rubber anti fatigue tile and cove base molding.
- i. Wall Panel: Exterior face of manufacturer's standard 16 gauge galvanized steel, E.P.S. insulation core and a 16 gauge galvanized steel interior face. Panels shall be MIG welded into place.
 - (1) Overall thickness: 3 inches R-12.

2.3 BUILDING ACCESSORIES

- a. Swinging Doors: 1-3/4 inches thick, tubular-frame design.
 - (1) Commercial Grade Steel Swing Door 36 inches by 84 inches with card reader, keyed entry and lever handle lockset. No Glass.
- b. Electrical: See electrical drawings.
- c. Thru-wall Heating/Air Conditioning: 208V
 - (1) 12,000/11,100 Heat Pump
- d. HVAC - Cooling And Heating (Mini Split System): The air handlers will be mounted on the wall/ceiling of the modular building. The compressors for the modular building HVAC shall be mounted on external pad to the building with all conduit and refrigerant lines to be properly sealed at all penetrations through the walls/roof.

2.4 FABRICATION

Fabricate factory built, prefabricated structures and shelters completely

in factory.

Separate dissimilar materials using nonconductive tape, paint, or other material not visible in finished work.

Fabricate factory built, prefabricated structures and shelters for forklift unloading under base of booth with forklift pockets in base of booth or welded in place or lifting lugs at roof that are suitable for placement of the structure on prepared foundations.

PART 3 EXECUTION

3.1 EXAMINATION

Examine supporting foundations for compliance with manufacturer's requirements, including installation tolerances and other conditions affecting performance of supporting members.

Check installed anchor bolts for accuracy. Verify that bearing surfaces are ready to receive the work.

Verify the rough-in of required mechanical and electrical services prior to placement of the structure.

If preparation is the responsibility of another installer, notify Architect of unsatisfactory preparation before proceeding.

3.2 PREPARATION

Clean surfaces thoroughly prior to installation.

Prepare surfaces using the methods recommended by the manufacturer for achieving the best result for the substrate under the project conditions.

3.3 INSTALLATION

Install in accordance with manufacturer's instructions.

Separate dissimilar materials using nonconductive tape, paint, or other material not visible in finished work.

Place on prepared concrete foundations and slabs provided as specified under Section 03 30 00.

Anchor securely in place, allowing for required movement, including expansion and contraction.

Connect mechanical services as specified under Division 23.

Connect electrical services as specified in Division 26.

3.4 PROTECTION

Protect installed products until completion of project.

Touch-up, repair or replace damaged products before Substantial Completion.

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